

(1)

For $m \geq 3$:

$$\zeta(m) - 1 = \sum_{n=m+1}^{\infty} \sum_{k=1}^{\infty} \sum_{l=1}^{\infty} \frac{1}{(k+l)^n} \quad (2)$$

For $\alpha, s > 1$:

$$\Gamma(s)\zeta(s, \alpha) = \int_0^{\infty} \frac{e^{-z(\alpha-1)} z^{s-1}}{e^z - 1} dz \quad (3)$$

For $s > 1, \alpha \geq 0$:

$$\Gamma(s)H_{\alpha,s} = \int_0^{\infty} \frac{1 - e^{-z\alpha}}{e^z - 1} z^{s-1} dz \quad (4)$$

For $s \geq 4$:

$$\frac{1}{2}\zeta(s-2) - \zeta(s-1) = -\frac{1}{2} + \sum_{n=s}^{\infty} \sum_{(m_1, m_2, m_3) \in \mathbb{Z}^3} (m_1 + m_2 + m_3)^{-n} \quad (5)$$

The next two functions are auxillary for the following identity.

$$Q(s) = \prod_{n=1}^{\infty} \prod_{m=1}^n \frac{1}{1 - (4n-1)^{-s} (4m-1)^{-s}}$$

Let $\{H_n\}$ be the Hilbert Primes.

$$T(s) = \prod_{n=1}^{\infty} \frac{1}{1 - H_n^{-s}}$$

Let $\{M_n\}$ be the sequence of primes congruent to 1 mod 4.

$$T(s) - Q(s) = \sum_{n=1}^{\infty} \frac{1}{M_n^s} \quad (6)$$

THM f has symmetric laurent expansion coefficients iff it's real on the complex unit circle

$$\sum_{n=0}^{\infty} \frac{1}{\left(n + \frac{1}{2}\right)^2} = \frac{\pi^2}{4} \quad (7)$$

$$\prod_{n=0}^{\infty} \left(1 - \left(\frac{x}{e^n} \right)^2 \right) = 1 + \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} \frac{(n+m-1)!}{n!} (-x^2)^m e^{-2n} \quad (8)$$

$$\zeta(m) - \zeta(m-1) = \sum_{n \in \mathbb{Z}^2} \frac{1}{(n_1 + n_2)^{m+1}} \quad (9)$$

$$\sum_{n=1}^{\infty} \sum_{k=n+1}^{\infty} \sum_{m=k+1}^{\infty} \frac{1}{(nmk)^2} = \frac{\pi^6}{5040} \quad (10)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \sum_{m=n+1}^{\infty} \frac{1}{m^2} = \frac{\pi^4}{36} \quad (11)$$

$$\int_{-\infty}^{\infty} e^{xt-e^t} dt = \Gamma(x) \quad (12)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \left(\frac{\pi^2}{3} H_{n,2} - H_{n,4} - H_{n,2}^2 \right) = \frac{\pi^6}{420} \quad (13)$$

$$\prod_{n=1}^{\infty} \left(1 - \frac{1}{1-4n^2} \right) = \frac{\pi}{2} \quad (14)$$

$$\prod_{n=1}^{\infty} \frac{\frac{1}{4}(1-16n^2)(2n-1)^2}{\left((2n-1)^2 - \frac{1}{4}\right)(1-4n^2)} = 2 \quad (15)$$

$$\prod_{n=1}^{\infty} \frac{n^2}{\left(n - \frac{1}{2}\right)^2} \frac{2n-1}{2n+1} = \frac{\pi}{2} \quad (16)$$

$$\int_0^1 \frac{1 - \cos(x \ln(1-t))}{t \ln(1-t)} dt = \frac{1}{2} \ln \left(\frac{2\pi x}{e^{x\pi} - e^{-x\pi}} \right) \quad (17)$$

$$\sum_{n=1}^{\infty} \sum_{m=n+1}^{\infty} \sum_{k=m+1}^{\infty} f(n) f(m) f(k) = \frac{S_1^3 - 3S_1S_2 + 2S_3}{6} \rightarrow \frac{2n^3 - 3n^2 + n}{2} \quad (18)$$

$$\sum_{n=1}^{\infty} \sum_{m=n+1}^{\infty} f(n) f(m) = \frac{S_1^2 - S_2}{2} \rightarrow \frac{n^2 - n}{2} \quad (19)$$

Where $S_k = \sum_n f(n)^k$.

$$\frac{10^{kn}}{(10^n - 1)^k} \quad (20)$$

gives all n -digit values in k th diagonal of pascal's triangle in order

$$\frac{10^{2n}}{10^{2n} - 10^n - 1} \quad (21)$$

gives all n -digit values for F_k in order

$$U_{-1} = \frac{-2\sqrt{x} \cot(\pi\sqrt{x})}{\pi} \quad (22)$$

$$U_{-2} = -\frac{4x}{\pi^2} - \frac{4\sqrt{x} \cot(\pi\sqrt{x})}{\pi^3} \quad (23)$$

$$2Q_{(-2)1} = \frac{-4}{\pi^4} \quad (24)$$

$$Q_{-n}Q_n = \frac{1}{n!(1-4n^2)} \quad (25)$$

$$-2 \cdot \frac{(2n-2)!}{(n-1)!\pi^{2n}} \quad (26)$$

$$\int_0^1 \frac{1 - \ln\left(\frac{1}{u}\right)^{x+1}}{1 + \ln(u)} du = \sum_{n=0}^x n! \quad (27)$$

$$\sum_{n=k}^m f^n(a) = e^a \int_a^\infty \left(f^{(k)}(x) - f^{(m+1)}(x) \right) e^{-x} dx; \quad (28)$$

$$\sum_{n=0}^{\infty} \frac{1}{(sn)!} = \frac{1}{s} \sum_{n=0}^{s-1} e^{e^{-\frac{2i\pi n}{s}}} = \frac{1}{s} \sum_{n=0}^{s-1} e^{\cos(\frac{2\pi n}{s})} \cos\left(\sin\left(\frac{2\pi n}{s}\right)\right) \quad (29)$$

$$-\frac{1}{2\pi} \int_{|z|=1} \frac{e^z}{\ln(z) - \ln(x)} \left(\psi_{\frac{x}{z}}(1) + \ln(z-x) - \ln(z) \right) dz = \quad (30)$$

$$\frac{1}{2\pi i} \int_{|z|=1} \frac{e^z}{z} \sum_{n=1}^{\infty} \frac{x^n}{z^n - x^n} dz = \sum_{n=1}^{\infty} d(n) \cdot \frac{x^n}{n!} \quad (31)$$

if $|x| < 1$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{e^{-i\theta}} f(xe^{i\theta}) d\theta = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \frac{x^n}{n!} \quad (32)$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{xe^{i\theta} + e^{-i\theta}} d\theta = \sum_{n=0}^{\infty} \frac{x^n}{n!^2} \quad (33)$$

$$\sum_{n=0}^{\infty} \frac{\cos(n)}{n!^2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{\cos(\theta+1) + \cos(\theta)} \cos(\sin(\theta) - \sin(\theta+1)) d\theta \quad (34)$$

$$-\frac{1}{2\pi^{\frac{3}{2}}} \int_0^{2\pi} \sqrt{\frac{3-e^{i\theta}}{1-e^{i\theta}}} e^{i\theta + \frac{2}{1-e^{i\theta}}} d\theta = \sum_{n=0}^{\infty} \frac{1}{n!(0.5-n)!(0.5+n)!} \quad (35)$$

$$\int_0^1 \cos\left(\sqrt{x \ln(s)}\right) ds = \sum_{n=0}^{\infty} \frac{n! x^n}{(2n)!} \quad (36)$$

$$\frac{1}{\pi} \int_{-1}^1 \frac{e^{2u}}{\sqrt{1-u^2}} du = \sum_{n=0}^{\infty} \frac{1}{n!^2} \quad (37)$$

$$\frac{1}{2\pi} \int_{-\frac{3}{4}\pi}^{\frac{5}{4}\pi} e^{\sqrt{2}\sin(x)} \cos\left(\sqrt{2}\sin(x)\right) dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!^2} \quad (38)$$

$$\int_0^1 I_m\left(\sqrt{-2x \ln(t)}\right) \ln^{-\frac{m}{2}}(t) dt = \frac{1}{(-2x)^{\frac{m}{2}}} \sum_{n=m}^{\infty} \frac{x^n}{n!} \quad (39)$$

$$-2 \int_0^1 \frac{e^{2i\pi u} e^{\frac{2}{1-e^{2i\pi u}}}}{4+x^2(1-e^{2i\pi u})^2} du = \frac{\sin(x)}{x} \quad (40)$$

$$\int_0^1 \frac{e^{\frac{2}{1-e^{2i\pi u}}}}{1-e^{-2i\pi u}} \cdot \arctan\left(\frac{x}{2}(1-e^{2i\pi u})\right) du = \text{Si}(x) \quad (41)$$

$$\int_0^1 f(-x \ln(t)) dt = \sum_{n=0}^{\infty} f^{(n)}(0) x^n \quad (42)$$

$$f(x) = \sum_{n=0}^{\infty} c_n x^{a_n} \implies -\frac{1}{2\pi} \int_0^{2\pi} f\left(\frac{x}{2}(1-e^{i\theta})\right) \frac{e^{\frac{2}{1-e^{i\theta}}}}{e^{-i\theta}-1} d\theta = \sum_{n=0}^{\infty} \frac{c_n x^{a_n}}{(a_n)!} \quad (43)$$

$$\int_0^1 \frac{1}{\sqrt{\ln\left(\frac{1}{x}\right)}} dx = \sqrt{\pi} \quad (44)$$

$$\forall \{c_n\}, \{a_n\} s.t. f = \sum_{n=0}^{\infty} c_n x^{a_n}, \sum_{n=0}^{\infty} c_n / (a_n)!^k \text{ is the same regardless of choice of specific } \{a_n\}, \{c_n\} \quad (45)$$

assuming $a_n \notin \mathbb{Z}_-, k \in \mathbb{Z}$

$$i \int_{i-\infty}^{i+\infty} \frac{q^{z^2} \cot(\pi z)}{((z^2)!)^k} dz = 1 + 2 \sum_{n=1}^{\infty} \frac{q^{n^2}}{(n^2)!^k} \quad (46)$$

$$\sum_{n=0}^{\infty} \frac{\sin(n^2)}{(n^2)!} = \frac{i}{2} \int_{i-\infty}^{i+\infty} \frac{\sin(z^2) \cot(\pi z)}{\Gamma(z^2 + 1)} du \quad (47)$$

$$\frac{1}{\sqrt{\pi}} \int_0^1 \frac{\sin(2\sqrt{\ln(t)})}{\sqrt{\ln(t)}} dt = \sum_{n=0}^{\infty} \frac{1}{(n + \frac{1}{2})!} \quad (48)$$

$$\frac{1}{\pi} \int_0^{\pi} \frac{e^{\cos(\theta)} \cos(\frac{\theta}{2} + \sin(\theta))}{\sqrt{\sin(\theta)}} d\theta = \sum_{n=0}^{\infty} \frac{1}{(n!)^2} 0.5^{2n} \quad (49)$$

$$\sum_{n=0}^{\infty} \frac{x^n}{n!^2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{e^{-i\theta}} \frac{1}{\sqrt{1 - 4x(e^{i\theta})^2}} d\theta \quad \left(|x| < \frac{1}{4}\right) \quad (50)$$

$$\sum_{n=0}^{\infty} \frac{\Gamma(s+n)}{n!^2} = \frac{\Gamma(s)}{2\pi} \int_{-\pi}^{\pi} \frac{e^{e^{-i\theta}}}{(1 - e^{i\theta})^s} d\theta \quad (51)$$

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{e^z}{z} \frac{1}{\zeta(-\ln(z))} dz = 2 \sum_{n=1}^{\infty} \frac{e^{e^{-2n}} (-1)^n (2\pi)^{2n}}{(2n)! \zeta(2n+1)} \quad (52)$$

$$\sum_{n=0}^{\infty} \frac{1}{\pi^2 (2n+1)^2 + 1} = \frac{e-1}{e+1} \frac{1}{4} \quad (53)$$

$$\sum_{k=1}^{\infty} \frac{x^k}{\zeta(k+1)} = x \sum_{n=1}^{\infty} \frac{\mu(n)}{n(n-x)} \quad (54)$$

$$A\{f\}(x) = L^{-1} \left\{ \frac{1}{x} f\left(\frac{1}{x}\right) \right\}(x) \quad (55)$$

$$A^{-1}\{f\}(x) = \frac{1}{x} L\{f\}\left(\frac{1}{x}\right) \quad (56)$$

$$\sum_{n=1}^{\infty} \zeta(2n) x^{2n} = -\frac{\pi x}{2} \cot(\pi x) + \frac{1}{2} \quad (57)$$

$$x^2 \sum_{n=1}^{\infty} \frac{\mu(n)}{n^2 - x^2} = \sum_{n=1}^{\infty} \frac{x^{2n}}{\zeta(2n)} \quad (58)$$

$$\sum_{n=1}^{\infty} \frac{x^{2n+1}}{\zeta(2n+1)} = x^3 \sum_{n=1}^{\infty} \frac{\mu(n)}{n(n^2 - x^2)} \quad (59)$$

$$\sum_{n=1}^{\infty} \frac{\mu(n)}{n} (e^{\frac{x}{n^2}} - 1) = \sum_{n=1}^{\infty} \frac{x^n}{n! \zeta(2n+1)} \quad (60)$$

$$\prod_{n=1}^{\infty} \frac{n^2}{n^2 + q} = \frac{2\pi\sqrt{q}}{e^{\pi\sqrt{q}} - e^{-\pi\sqrt{q}}} \quad (61)$$

$$\prod_{m=1}^{\infty} \frac{me^{\frac{1}{m}}}{m+1} = e^{\gamma} \quad (62)$$

$$\sum_{k=1}^{\infty} \frac{x^{nk} \zeta(nk)}{k} = \sum_{k=1}^n \ln \left(\Gamma \left(1 - xe^{\frac{2i\pi k}{n}} \right) \right) \quad (|x| < 1) \quad (63)$$

$$\sum_{n=0}^{\infty} \frac{n!}{(2n)!} = \frac{e^{\frac{1}{4}} \sqrt{\pi}}{2} \operatorname{erf} \left(\frac{1}{2} \right) + 1 \quad (64)$$

$$\ln(\Gamma(1-z)) = \gamma z + \sum_{k=2}^{\infty} \frac{\zeta(k)}{k} z^k \quad (65)$$

$$\sum_{n=2}^{\infty} \frac{\zeta(n)}{n^s} = \int_0^{\infty} x^{s-2} \ln(\Gamma(1-e^{-x})) dx - \gamma \quad (66)$$

$$\frac{\sqrt{x} - \sqrt{x-1}}{\zeta\left(\frac{1}{2}\right)} + \sum_{\rho} \frac{(x^{\rho} - (x-1)^{\rho}) \zeta(2\rho)}{\rho \zeta'(\rho)} = \lambda(x) \quad (67)$$

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(z) \zeta(az)}{(-\ln(x))^z} dz = \sum_{n=1}^{\infty} x^{n^a} \quad (68)$$

$$F(s) = \sum_{n=1}^{\infty} \frac{C_n}{n^s} \rightarrow \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(z)}{(-\ln(x))^z} F(az) dz = \sum_{n=1}^{\infty} C_n x^{n^a} \quad (69)$$

$$\Gamma(s) \zeta(s+1) = - \int_0^{\infty} x^{s-1} \ln(1-e^{-x}) dx \quad (70)$$

$$\sum_{m=2}^{\infty} \frac{\Lambda(m)}{\ln(m)} \sum_{n=1}^{\infty} \frac{\mu(n)}{n} x^{m^n} = \sum_p x^p \quad (71)$$

$$f(x) = \sum_{n=0}^{\infty} e^{-x^n} \implies k \text{ is prime iff } \lim_{x \rightarrow 0} f^{(k)}(x) = \pm k! \quad (72)$$

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(z) x^z}{x^z - 1} dz = \sum_{n=0}^{\infty} e^{-x^n} \quad (73)$$

The next two do not converge in the usual sense:

$$\pi'(e^x) = \left(\sum_{n=1}^{\infty} \frac{x^n}{\zeta(n+1)n!} \right) \odot \left(\frac{x}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\ln(\zeta(z))}{1-xz} dz \right) \quad (74)$$

$$\pi'(x) = \frac{1}{2\pi i} \sum_{n=1}^{\infty} \frac{\ln(x)^n}{n! \zeta(n+1)} \int_{c-i\infty}^{c+i\infty} \ln(\zeta(z)) z^{n-1} dz \quad (75)$$

$$\pi'(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\ln(\zeta(z))}{z} \sum_{n=1}^{\infty} \frac{\mu(n)}{n} x^{\frac{z}{n}} dz \quad (76)$$

$$\pi'(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\ln(\zeta(z))}{z} \sum_{n=1}^{\infty} \frac{(z \ln(x))^n}{n! \zeta(n+1)} dz \quad (77)$$

$$\sum_{n=1}^{\infty} \frac{\zeta(2n+1)}{(2n+1)4^n} = \ln(2) - \gamma \quad (78)$$

$$\sum_{n=1}^{\infty} \frac{\zeta(2n)}{n \cdot 4^n} = \ln\left(\frac{\pi}{2}\right) \quad (79)$$

$$2 \sum_{n=2}^{\infty} \frac{\zeta(n)}{2^n \cdot n} + \gamma = \ln(\pi) \quad (80)$$

$$\prod_{m=1}^{\infty} \left(\frac{2m-1}{2m} \right)^2 e^{\frac{1}{m}} = \frac{e^{\gamma}}{\pi} \quad (81)$$

$$\prod_{m=1}^{\infty} \frac{4m^3 - 3m + 1}{4m^3} = \frac{1}{\pi} \quad (82)$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{2x \cos(\theta)} d\theta = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!^2} \quad (83)$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{1 - 2\sqrt{x} \cos(\theta)} d\theta = \sum_{k=0}^{\infty} \frac{(2k)!}{(k)!^2} x^k \quad (84)$$

$$\Gamma(s) \sum_{(m,n) \in \mathbb{Z}^2} \frac{1}{(m^a + n^a)^s} = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(z) \Gamma(s-z) \zeta(az) \zeta(a(s-z)) dz \quad (85)$$

$$\sum_{n=0}^{\infty} x^{q^n} = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(z)}{(q^{-z} - 1)(-\ln(x))^z} dz \quad (86)$$

$$F_k(a, s) = \Gamma(s) \sum_{n \in \mathbb{Z}^k} \frac{1}{(n_1^a + n_2^a + \dots + n_k^a)^s} \implies F_k(a, s) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F_1(a, z) F_{k-1}(a, s-z) dz \quad (87)$$

$$\int_0^{\infty} \frac{|\zeta(\frac{a}{2} + ait)|^2}{\cosh(\pi t)} dt = \sum_{(n,m) \in \mathbb{N}^2} \frac{1}{(n^a + m^a)} \quad (88)$$

$$\frac{(2\pi)^3}{(1 - e^{-2\pi})(1 - 2\cos(\pi\sqrt{3})e^{\pi} + e^{2\pi})} = \prod_{n=1}^{\infty} \frac{n^6}{n^6 + 1} \quad (89)$$

$$\prod_{n=1}^{\infty} \frac{n^{4m+2}}{n^{4m+2} - 1} = (2\pi)^{2m+1} \prod_{k=1}^{2m+1} \frac{e^{\pi e^{\frac{i\pi k}{2m+1}}}}{e^{2\pi e^{\frac{i\pi k}{2m+1}}} - 1} \quad (90)$$

$$\Gamma(s) \sum_{m,n=1}^{\infty} \frac{1}{(m^a + n^a)^s} = \int_0^{\infty} \left| \Gamma\left(\frac{s}{2} + it\right) \zeta\left(\frac{as}{2} + ait\right) \right|^2 dt \quad (91)$$

$$\Gamma(s) \sum_{m,n=1}^{\infty} \frac{1}{(m^a + n^a)^s} = \left\| \Gamma\left(\frac{s}{2} + it\right) \zeta\left(\frac{as}{2} + ait\right) \right\|_2^2 \quad (92)$$

$$\sum_{n=1}^{\infty} a_n x^{\frac{1}{n}} = \sum_{n=0}^{\infty} \frac{\ln(x)^n}{n!} \zeta_A(n) \quad (93)$$

$$c(n) = \sum_{m=1}^{n-1} a_m a_{n-m} \implies \pi \Gamma(s) \sum_{n=1}^{\infty} \frac{c_n}{n^s} = \int_0^{\infty} \left| \Gamma\left(\frac{s}{2} + it\right) \zeta_A\left(\frac{s}{2} + it\right) \right|^2 dt \quad (94)$$

$$\sum_{n=0}^{\infty} \frac{1}{2n+1} \frac{(-1)^n}{n! (0.5-n)!} = \frac{\sqrt{\pi}}{2} \quad (95)$$

$$\|\Gamma(\frac{s}{2} + it)\|_2^2 = \int_0^{\infty} \left| \Gamma\left(\frac{s}{2} + it\right) \right|^2 dt = \pi 2^{-s} \Gamma(s) \quad (96)$$

$$\sum_{\rho} \frac{1}{\rho} = \ln\left(\frac{e^{1+\frac{\gamma}{2}}}{2\sqrt{\pi}}\right) = \sum_{\rho} |\rho|^{-2} \quad (97)$$

For any completely multiplicative function λ defined on $\mathbb{N}^*$, where $f(x) = \sum_n \lambda(n)x^n$

$$\lambda(s) = \frac{1}{f(b^{-s})} \sum_{n=1}^{\infty} \frac{\lambda(n)}{b^n} \cdot \frac{1}{2\pi i} \int_{|z|=a} \frac{z^{s+n-1}}{z^s - 1} dz \quad (b > a > 1) \quad (98)$$

$$s\psi(-s) - 1 = \sum_{m=1}^{\infty} \zeta(m) s^m \quad (99)$$

$$\sum_{n=1}^{\infty} \frac{\sin(nq)}{n^s} = \frac{\sin(q)}{2\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{\cosh(x) - \cos(q)} dx \quad (100)$$

$$\int_0^{\infty} \frac{1}{2 \cosh(t) - 1} dt = \frac{2\pi}{3\sqrt{3}} \quad (101)$$

$$\sum_{n=1}^{\infty} \frac{(2n)!}{n! (n+1)!} \frac{4^{-n}}{2n+3} = \frac{5}{3} - \frac{\pi}{2} \quad (102)$$

$$\frac{1}{2} \int_0^{\infty} \frac{t}{\cosh(t)} dt = C \quad (103)$$

$$\sum_{n=0}^{\infty} \frac{(n - \frac{3}{2})!}{n! 2^n} = -\sqrt{2\pi} \quad (104)$$

$$\int_0^{\infty} \frac{1}{\cosh(t) - x} dt = \frac{\pi + i \sinh^{-1}(x)}{\sqrt{1-x^2}} \quad (105)$$

$$\int_{-\pi}^{\pi} \frac{1}{\sqrt{2} - \cos(\theta)} d\theta = 2\pi \quad (106)$$

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{1}{\zeta(s) \sin(\pi s)} ds = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{\mu(n)}{n+1} \quad (107)$$

$$\sum_{n=0}^{\infty} \frac{\mu_2(2n+1)}{(2n+1)^2} = \frac{8}{\pi^2} \quad (108)$$

$$\zeta_2(s) = \frac{2}{\zeta(s) + \eta(s)} \quad (109)$$

$$\sum_{p \equiv a \pmod m} x^p = \frac{1}{m} \sum_{k=1}^m e^{\frac{-2i\pi k a}{m}} \cdot \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \left(-\ln \left(x e^{\frac{2i\pi k}{m}} \right) \right)^{-s} \Gamma(s) \zeta_P(s) ds \quad (110)$$

$$\pi(x) = {}^R \sum_{n=1}^{\infty} \frac{\ln(x)^n}{n! n \zeta(n+1)} ((-2)^n \zeta(-n) + 1) \quad (111)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^s (n+q)} = \frac{1}{\Gamma(s)} \int_0^{\infty} x^{s-1} (2e^{-xq} \ln(1-e^{-x}) + \phi(e^{-x}, 1, q)) dx \quad (112)$$

$$\ln(x\#) = -\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\zeta'(u)}{\zeta(u)} \cdot \frac{1}{u} \sum_{n \geq 1} \mu(n) x^{\frac{u}{n}} du \quad (113)$$

$$= {}^R \frac{1}{2\pi i} \sum_{n=1}^{\infty} \frac{(n-1) \ln(x)^n}{n! \zeta(n+1)} \int_{c-i\infty}^{c+i\infty} \ln(\zeta(u)) u^{n-2} du \quad (114)$$

$$|\{p \in \mathbb{P} \mid p+k \in \mathbb{P}\}| = \frac{-1}{4\pi^3} \int_{c-i\infty}^{c+i\infty} \int_{c-i\infty}^{c+i\infty} \Gamma(z_1) \zeta_p(z_1) \Gamma(z_2) \zeta_p(z_2) \frac{\sin(\pi(z_1 - z_2) + \pi k)}{z_1 - z_2 + k} dz_2 dz_1 \quad (115)$$

$$\pi(x) = \frac{-1}{2\pi^2 i} \int_{c-i\infty}^{c+i\infty} \Gamma(z) \zeta_p(z) (\phi(-1, 1, z+x+2) \sin(\pi(z+x)) + \phi(-1, 1, z+1) \sin(\pi z)) dz \quad (116)$$

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(z) \zeta_a(z+s) (-\ln(x))^{-z} dz = \sum_{n=1}^{\infty} \frac{a_n x^n}{n^s} \quad (117)$$

$$\ln(x\#) = \frac{-1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\zeta'(u)}{\zeta(u)} \sum_{m=2}^{\infty} \frac{\ln^m(x)}{\zeta(m) m!} u^{m-1} du \quad (118)$$

$$\sum_p x^p = {}^R \frac{1}{2\pi i} \sum_{m=0}^{\infty} \ln(-\ln(x))^m \frac{(-1)^m}{m!} \sum_{k=-1}^{\infty} \frac{\Gamma_k}{\zeta(m+k+2)} \int_{c-i\infty}^{c+i\infty} \ln(\zeta(z)) z^{m+k} dz \quad (119)$$

$$\begin{aligned} c_1 \psi_{xx} + c_2 \psi &= \psi_t \\ \psi(x, 0) &= g(x) \\ g(x + 2\pi) &= g(x) \end{aligned}$$

$$\implies \psi(x, t) = \frac{e^{tc_2}}{2\pi} \int_{-\pi}^{\pi} \vartheta\left(\frac{x-\theta}{2\pi}; \frac{c_1 ti}{\pi}\right) g(\theta) d\theta \quad (120)$$

$$\int_0^{2\pi} \sqrt{\frac{3-e^{i\theta}}{1-e^{i\theta}}} e^{i\theta + \frac{2}{1-e^{i\theta}}} d\theta = -6\pi \operatorname{erf}(1) - 4\frac{\sqrt{\pi}}{e} \quad (121)$$

$$\pi e = \sum_{k=1}^{\infty} \frac{(e^2-1) \cos(1-2k) - (1+e^2)(2k-1) \sin(1-2k)}{2k(k-1)+1} \quad (122)$$

$$x = \sum_{k=0}^{\infty} \frac{(-1)^k \mu(2k+1)}{2k+1} \arctan(x^{2k+1}) \quad (123)$$

$$\frac{4}{\pi} = \sum_{k=0}^{\infty} \frac{(-1)^k \mu(2k+1)}{2k+1} \quad (124)$$

$$\frac{e^{\frac{1}{4}}}{2\pi i} \int_{-\pi}^{\pi} e^{-\frac{i\theta}{2} + e^{-i\theta}} \int_{(-4e^{i\theta})^{-\frac{1}{2}}}^{\infty} e^{-z^2} dz d\theta = \sum_{n=0}^{\infty} \frac{(2n)!}{n!^2} \quad (125)$$

$$\frac{\pi}{4} - \frac{1}{2} \ln(2) - \sum_{k=2}^n 2^{-k} (1-2^{1-k}) \zeta(k) = \int_{[0,1]^n} \arctan\left(\prod_{i=1}^n x_i\right) d\vec{x} \quad (126)$$

$$\sum_{k=2}^{\infty} \frac{\zeta(k)}{4^k} = \frac{3}{4} \ln(2) - \frac{\pi}{8} \quad (127)$$

$$\zeta(s) = \frac{2^s}{1-2^{1-s}} \int_0^1 \arctan(x) \frac{(-\ln(x))^{s-2}}{(s-2)!} \left(1 + \frac{\ln(x)}{s-1}\right) dx \quad (128)$$

$$\int_{[0,1]^s} f\left(\prod_{i=1}^s x_i\right) d\vec{x} = \int_0^1 f(x) \frac{(-\ln(x))^{s-1}}{\Gamma(s)} dx \quad (129)$$

$$\sum_{n=1}^{\infty} \int_{[0,1]^n} f\left(\prod_{i=1}^n x_i\right) d\vec{x} = \int_0^1 \frac{f(x)}{x} dx \quad (130)$$

$$\sum_{n=1}^{\infty} \int_{[0,\infty]^n} f\left(\sum_{i=1}^n u_i\right) du = \int_0^{\infty} f(u) e^u du \quad (131)$$

This implies that for any real $x > 0$, the number of real positive tuples which add to x is proportional to e^x .

For a fixed n , the number of n -tuples which add to x is proportional to $x^{n-1}/(n-1)!$.

$$x^s \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s - x^s} = \sum_{m=1}^{\infty} \frac{x^{sm}}{\zeta(sm)} \quad (132)$$